



Associations between vertical and horizontal jumps with power values during deadlift exercise among youth soccer (football) players

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ABSTRACT:

Objectives: The similarity in joint angles and performance between the deadlift and vertical jumps was previously displayed. Additionally, the hexagonal barbell produces greater peak power compared to performance with traditional Olympic barbell. To optimally develop power and performance in explosive tasks in sport (such as jumps), training with the load that maximizes the power output is considered ideal. Therefore, the objective of this research is to investigate the relationship between jumps and power values during the deadlift exercise performed with a hexagonal barbell.

Methods: 14 youth soccer players (16-18 years old) were tested on the Sargent jump test, broad jump test, one repetition maximum (1RM) using the PUSH 2.0 band that was used for displaying the power values at 45, 55, 65, 75, 85, and 95% of 1RM. Correlations between jumps and power values at different percentages of 1RM were evaluated by Pearson's correlation coefficients.

Results: The significant correlation ($p < 0.05$) was evidenced between Sargent jump test and power value at 45, 55, 65, 75% 1RM ($r = 0.75$; $r = 0.60$; $r = 0.56$; $r = 0.53$, respectively), and between Broad jump and power at 45, 55, 65% 1RM ($r = 0.71$; $r = 0.51$; $r = 0.54$, respectively).

Conclusion: Power values at moderate loads were significantly correlated with vertical and horizontal jumps. Therefore, it could be expected that training at moderate loads that produce the greatest power values will lead to greater development of jumping performance, which can be important also for improving soccer performance.

Keywords: power, performance, team sports, adolescents



INTRODUCTION

Strength training or resistance training involves performing physical exercises designed to improve strength and power. Recently, one of the popular resistance training methods is Velocity Based Training (VBT) (1). VBT is a training method based on the velocity of movement of the load, where daily oscillations in performance can be resolved by adjusting the load in training (2). This method is based on the technology such as accelerometers (e.g., PUSH Band) or linear position transducers (e.g. GymAware) used to measure movement speeds during exercise (e.g., deadlift) (3). Using linear position transducers and accelerometers, the load velocity is calculated (i.e., bar lifting speed during deadlift) and a load velocity profile can be created (4). Also, it provides coach and athlete immediate feedback on the performance of the movement/exercise (1).

Devices used for VBT measure different values during the movement, including movement velocity, force, and power (3). One of the most important values is power output, which has a significant role in sports performance. More precisely, research suggests that improvements in power output lead to significant improvements in sports performance (5). Power is defined as “the rate of performing work, the product of force and velocity, or the rate of transformation of metabolic potential energy to work or heat” (6). It has to be highlighted that “maximal muscular power is obtained at optimal values of force and velocity because power is the product of these 2 parameters” (7). Accordingly, VBT has displayed as efficient at improving power output during certain exercises and movements.

Previous research showed that training at different ranges of loads (percentages of 1RM) produces different (lower or greater) power outputs. Namely, results obtained by

Baker, Nance (8) showed that loads at 55-59% of 1RM during the jump squat exercise provoked the greatest power outputs (8). Additionally, Soriano, Jiménez-Reyes (9) suggested that moderate loads at 30-70% of 1RM represent the optimal load for power production during the squat exercise. Hence, previous studies have suggested that low to moderate loads elicit the greatest power values, which could be of practical significance in the way of developing performances in explosive movements that are often used in sports (i.e., jumping, throwing).

It has been hypothesized that particular strength-training exercises can produce significant improvements in sports performance tasks. Precisely, it was recorded that squat jump exercise led to improvements in vertical jumping (10). The authors of that study explained it by the biomechanical similarity of those two movements. Precisely, performing a squat jump in training led to increased neuromuscular efficiency and resulted in improvements in vertical jumps (10). In that manner, deadlift exercise with the hexagonal barbell has similar joint angles with the performance of jumps (11). Therefore, it could be assumed that deadlift training will lead to improved jumping performances.

However, there is a lack of studies that investigated the correlations between jumping performance and values recorded during the VBT while performing deadlift exercise. Therefore, the objective of this research is to investigate the relationship between jumps and power values during the deadlift exercise performed with a hexagonal barbell. The hypothesis of this study is that power values at low to moderate loads will be more correlated with jumping performances. The results of this study might be useful for prescribing the optimal load for developing jumping performances.

METHODS

Participants

This study included 14 young football players who were involved in at least 2 years of resistance training. Participants signed a written informed consent form after being informed of the purpose and experimental procedures, before the investigation began.

Variables

This research consisted of the following variables: 1RM measured by the wearable accelerometers (used for VBT monitoring and prescription) – VBT1RM, Sargent jump test (SJ), Broad jump test (BJ), and anthropometric values: Body height (BH), Body weight (BW), and body fat percentage (BF%).

The SJ was used to estimate vertical explosive power. Participants stood with both feet shoulder-width apart, and the maximum one-arm vertical reach was recorded. A vertical jump with counter-movement was performed and the reach was recorded by leaving a chalk-trace on the dark wall. The result of the jump was calculated by subtracting the height of the

reach from the recorded height of the jump. Subjects performed three jumps and the best value was included in the analysis.

The BJ was used to estimate horizontal explosive power. The participants were standing at the beginning of the line placed on the ground and jumped as far as they could. The distance from the starting line represented the result on the given test. The best result from the three jump attempts was included in the analysis.

One repetition maximum (VBT1RM) using the PUSH 2.0 band was used for displaying the power values at 45, 55, 65, 75, 85, and 95% of 1RM., Subjects were instructed to maintain eccentric control, before creating maximum force during the concentric phase at different percentages for all repetitions. Each series of repetitions during the protocol was interrelated with 3-5 minutes of rest.

Statistical analyses

The normality of all variables was checked by Kolmogorov Smirnov test for normality prior to all analyses. Correlations between jumps and power values at different percentages of 1RM were evaluated by Pearson's correlation coefficients. The level of statistical significance was set on $p < 0.05$.

STATISTICAL RESULTS

Table 1.
Descriptive statistics

Variable	Mean	St.Dev.	Minimum	Maximum
BH	181.25	5.99	168.00	188.00
BW	67.12	7.14	47.60	78.00
BF%	14.61	2.27	10.50	18.20
45AP	1087.59	306.92	691.73	1740.57
45PP	2479.42	577.34	1784.62	3727.47
55AP	1057.67	233.60	680.57	1481.42
55PP	2518.59	605.44	1607.50	3930.29
65AP	1050.37	330.42	733.74	1830.93
65PP	2191.91	845.95	0.81	3907.97
75AP	845.24	331.22	474.25	1588.89
75PP	1927.89	710.98	1082.53	3321.17
85AP	863.98	247.15	532.38	1303.49
85PP	1936.52	668.20	963.77	3078.32
95AP	729.56	212.87	375.78	994.01
95PP	1558.45	527.04	927.70	2580.53
SJ	47.00	5.66	40.00	56.50
BJ	225.15	15.96	201.00	247.00

Legend: BH – Body height, BW – Body weight, BF% - Body fat percentage, AP - Average power at 45, 55, 65, 75, 85, 95% of 1RM, PP – Peak power at 45, 55, 65, 75, 85, 95% of 1RM, SJ – Sargent jump test, BJ – Broad jump test

Table 1 shows results of descriptive statistics for all variables that are included in the study. Even from this analysis, it is visible that average and peak power values are higher at lower and moderate loads (percentages of 1RM).

Table 2.
Pearson product–moment correlation coefficients between jumps and power values at different percentages of 1RM

Variable	SJ	BJ
45AP	0.75*	0.71*
45PP	0.73*	0.70*
55AP	0.60*	0.51*
55PP	0.62*	0.54*
65AP	0.56*	0.54*
65PP	0.64*	0.58*
75AP	0.53*	0.42
75PP	0.65*	0.44
85AP	0.26	0.42
85PP	0.21	0.29
95AP	0.04	0.21
95PP	0.36	0.38

Legend: SJ – Sargent jump test, BJ – Broad jump test, AP - Average power at 45, 55, 65, 75, 85, 95% of 1RM, PP – Peak power at 45, 55, 65, 75, 85, 95% of 1RM, * level of significance ($p < 0.05$)

Table 2. displays Pearson product–moment correlation coefficients between jumps and power values at different percentages of 1RM. According to the results in this table, significant

correlations ($p < 0.05$) were evidenced between Sargent jump test and power values (average power) at 45, 55, 65, 75% 1RM ($r = 0.75$; $r = 0.60$; $r = 0.56$; $r = 0.53$, respectively), and between broad jump and power (average power) at 45, 55, 65% 1RM ($r = 0.71$; $r = 0.51$; $r = 0.54$, respectively).

DISCUSSION

The most important finding of this study is that power values during the deadlift exercise at moderate loads were significantly correlated with performances of vertical and horizontal jumps. Therefore, our initial study hypothesis has been accepted.

To explain such findings, results of previous studies investigating optimal power values at different loads will be presented. Precisely, Baker et al. (2001) found at rugby league players that moderate loads in the range of 48-63% of 1RM at full squat produced the highest power values (8). Also, Soriano, Jiménez-Reyes (9) suggested that moderate loads (30-70% of 1RM) represent the optimal load for power production during the squat exercise. Accordingly, Cormie, McCaulley (12) established that the peak power is maximized at 56% during the squat exercise (12). Similarly, Newton, Murphy (14) recorded that maximal power output occurred at loads of 30 and 45% of 1RM during bench press throws (14). Furthermore, Ramírez, Núñez (13) found that performing half-squat with a moderate load (65% of 1RM) led to an improvement in power output during the concentric phase of the squat among football players (13).

Collectively, previously mentioned studies suggested that performing exercises at low to moderate loads produce maximal power values. Therefore, it can be hypothesized that training with low to moderate loads would lead to the optimal development of

explosive exercises/movements (5). However, only few studies examined the effects of training at different loads on jumping performances. Precisely, de Hoyo, Gonzalo-Skok (15) found that 32 adolescent (under 19 years old) football players made significant progress in the vertical jump after 8 weeks of a back squat with moderate loads (40–60% 1RM). A similar finding was found by González-García, Morencos (16) for performing hip thrust on 24 adolescent girls in soccer players. They concluded that performing hip thrust at a moderate load (60% 1RM) leads to an improvement in the vertical component of the jumps (16). This was confirmed by Orange, Metcalfe (17) who displayed that loads at 40 and 60% of 1RM (representing optimal range for power output) led to improvements in jumping performance among rugby players (17).

The explanation of such findings could be found in the fact that movement velocities increase with decreased load, and that movement velocities decrease with increased loads. Both of these mechanisms lead to decreased power output (8). Therefore, the results of our study confirmed findings from previous studies, that is, power output is most likely maximized at moderate loads (in the range from 45 to 65% of 1RM). Put together, previous studies confirmed that performing exercises with low to moderate loads may lead to an improved adjustment in jumping performance. Also, a similar movement used in training and in testing itself is suitable for increasing neuromuscular efficiency, allowing the transmission to other biomechanically similar movements, ultimately resulting in favourable adjustments. However, even though our study is cross-sectional, it can be suggested that deadlift training at moderate loads would lead to improved jumping performances among youth soccer players.



CONCLUSION

In conclusion, it could be expected that training at moderate loads that produce the greatest power values will lead to greater development of jumping performance. This can be important for improving sports performance in general, including soccer performance. However, this study included youth players which means that findings should not be generalized on other populations and athletes. Therefore, future studies should investigate the correlations of values recorded during VBT and explosive sports tasks (i.e., jumps, throws, sprints) among older athletes and at other kinds of sports.

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